	(0.90)	
6(a)	The e.m.f. induced in a conductor is directly proportional to the rate of change of magnetic flux linkage	B1
(b)(i)	Any two of $t_1, t_3, t_5, t_7.$ Where gradient of the current-time graph is zero,	M1
	Therefore, rate of change of magnetic flux linkage is zero, so no e.m.f. is induced. $\frac{d(N\Phi)}{dt} = 0,$	A1
(b)(ii)	Between t_1 and t_2, the current in the solenoid is **decreasing.** Hence, the magnetic flux linkage through coil C is decreasing.	M1
	By **Lenz's law**, the induced magnetic field opposes this decrease by reinforcing the original magnetic field. Hence, it would be in the **same direction** as the original magnetic field.	A1
(c)	Area $A = \pi r^2 = \pi(1.8 \times 10^{-2})^2 = 1.018 \times 10^{-3}$ m^2 Change in magnetic flux density $\Delta B = 9.4 \times 10^{-4} \times (5 - (-5)) = 9.4 \times 10^{-3}$ T	M1
	Change in flux linkage $\Delta(N\Phi) = NA\Delta B = 63(1.018 \times 10^{-3})(9.4 \times 10^{-3}) = 6.03 \times 10^{-4}$ Wb turns	M1
	Average induced e.m.f. $\mathcal{E}_{\text{avg}} = \frac{\Delta(N\Phi)}{\Delta t} = \frac{6.03 \times 10^{-4}}{6.0 \times 10^{-3}} = 0.100$ V	A1